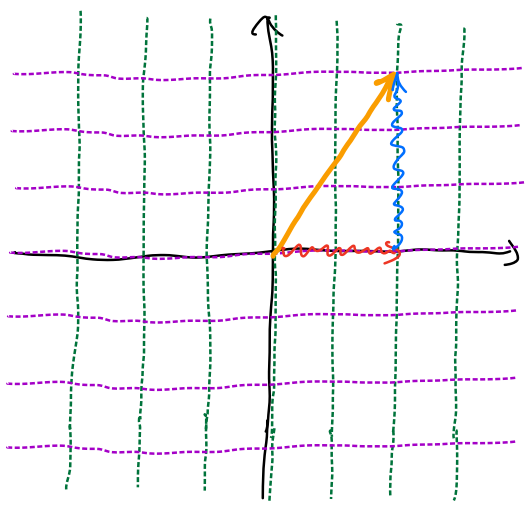
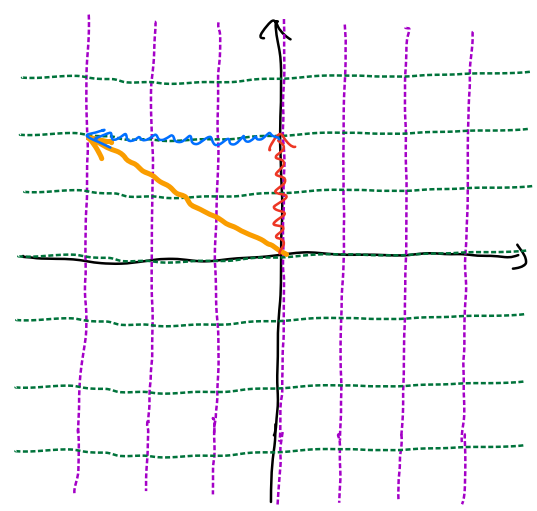
Unit 2: Linear Transformations

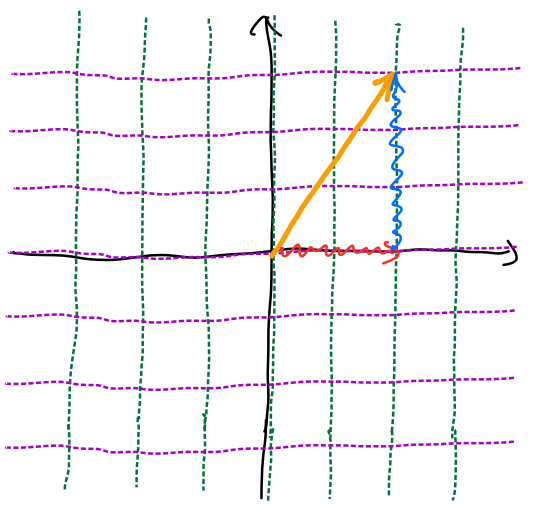
This unit is about "special" fcns btwn v.s.

Motivation: So far, we studied properties of v.s.
What about fcns/maps btwn v.sps?
Look at some photoshop operations:

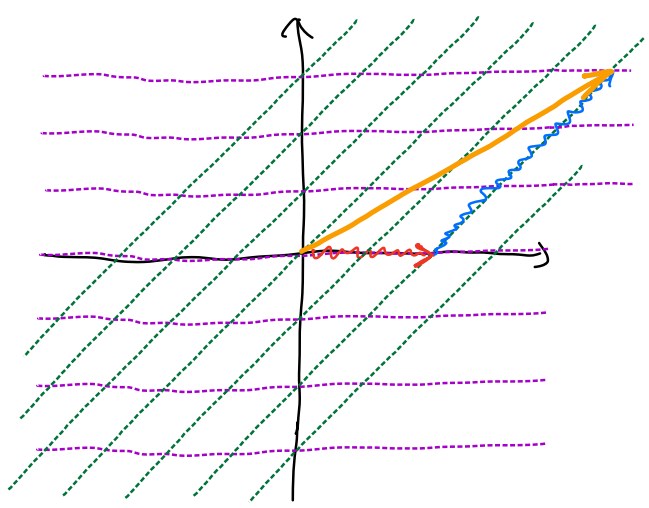


rotation by $\frac{\pi}{2}$ ↺





shear $\square \mapsto$



They fix origin, send any line to a line, and send any parallelogram to a parallelogram. Using  , we want fcns that preserve addition & scalar-mult.

Set Theory Warm-Up: given a fcn/map $f: X \to Y$,
(domain: X; codomain: Y)

$\operatorname{im}(f) = \underset{\text{(FIS)}}{\text{range}}(f) = f(X) = \{f(x) \in Y : x \in X\} \subseteq Y$ is called

range/image (im) of f. Given $Z \subseteq Y$,

$f^{-1}(Z) := \{x \in X : f(x) \in Z\}$ is the preimage (preim) of $Z \subseteq Y$ in X.

- when f is 1-1 & onto, $f^{-1}: Y \to X$ exists, and
- $f^{-1}(Z)$ is the img of $Z \subseteq Y$ via f^{-1}.

§2.1 Linear Maps: ker, im, and isom

As in Motivation, let's consider fcns btwn vector spaces that preserve $+$ & $\cdot$:

Def 2.1: V, W v.s. A fcn map $T: V \to W$ is a linear transformation (lin transf) (or lin map / homomorphism (hom)) if $\forall v_1, v_2 \in V$,

(1) $T(v_1 +_V v_2) = T(v_1) +_W T(v_2)$ $\left(T(v_1 + v_2) = T(v_1) + T(v_2)\right)$

(2) $T(c \cdot_V v_1) = c \cdot_W T(v_1)$ $\forall c \in \mathbb{R}$. $\left(T(cv_1) = c\,T(v_1)\right)$

The set of lin maps $V \to W$ is called $\mathcal{L}(V, W)$.

(When $V = W$, we say $\mathcal{L}(V) := \mathcal{L}(V, V)$)

Rmk 2.2: $T \in \mathcal{L}(V, W)$. Then

(1) $T(0_V) = 0_W$ $\quad$ $(T(0) = 0)$

(2) $T(-v) = -T(v) \quad \forall v \in V$

Pf: (1): Fix any $v \in V$. Then

$$T(0_V) = T(0 \cdot_V v) \overset{\text{Def 2.1}}{=} 0 \cdot_W T(v) \overset{\text{Prop 1.17(1)}}{=} 0_W. \qquad \blacksquare$$

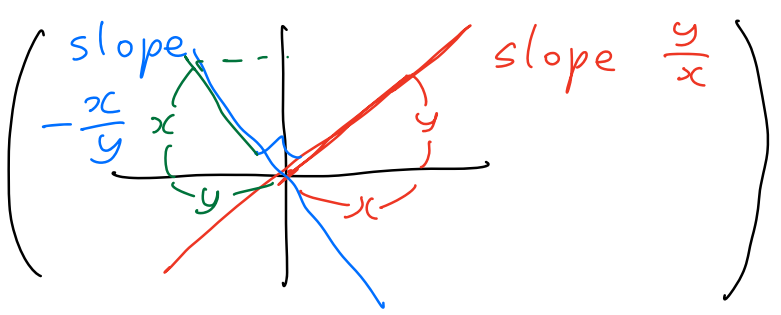
(2): $\forall v \in V$, $T(-v) = T((-1) \cdot_V v) \overset{\text{Def 2.1}}{=} (-1) \cdot_W T(v) \overset{\text{Prop 1.17(2)}}{=} -T(v).$ $\qquad \square$

Ex 2.3:

(1) Rotation by $\frac{\pi}{2}$ CCW in $\mathbb{R}^2$ is def by

$$R : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \longmapsto \begin{bmatrix} -y \\ x \end{bmatrix}$$



This is a lin map.

Pf: